

TactPrint: 3D Printing Lattice-based Tactile Displays with Optimized and Local Vibration

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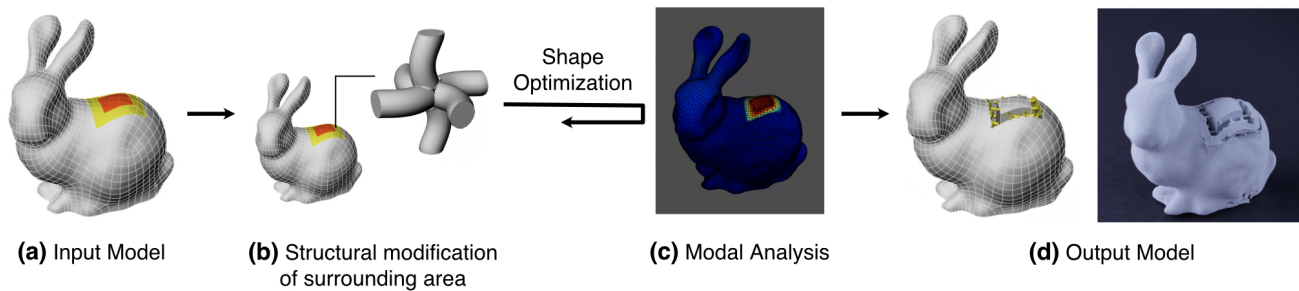


Figure 1: The design process of TactPrint: (a) the designers and prototypers input a 3D model and their desired local vibration region (red). (b) Then, TactPrint modifies the structure of the surrounding area of the region (yellow) to a sinusoidal-shape lattice structure, and (c) optimizes their structural shape using modal analysis. (d) Finally, TactPrint outputs the converted model and the designers and prototypers print out the model using a 3D printer.

ABSTRACT

Tactile sensation is widely used in human-object interactions. However, designing local vibrations within a device’s enclosure, while considering the propagation characteristics of the vibrations, is still challenging, which often results in the entire device vibrating uniformly. In this paper, we introduce TactPrint, a method that involves embedding a lattice structure around the area where we want to induce local vibrations and then optimizing this 3D structure to create a tactile display with the desired local vibration characteristics. With the TactPrint system, users can obtain structures with desired local vibration characteristics simply by inputting a 3D model, which can be fabricated with a 3D printer and combined

with an actuator to create a tactile display easily. In this demonstration, we present several prototypes of tactile displays created using our proposed system and application scenarios using them.

CCS CONCEPTS

• Applied computing → Computer-aided design; • Human-centered computing → Haptic devices.

KEYWORDS

local vibration presentation, lattice structure, computational design and fabrication

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1 INTRODUCTION

Many of our everyday devices are enriched by tactile sensations such as Haptic Touch by Apple, which works as important feedback

for human-object interaction. However, such tactile feedback is typically limited to a single vibration frequency and a single vibration region (i.e., the whole body of the device), rather than spatially distributed or separated vibrations with multiple frequencies. This limitation comes from the fact that it is currently difficult for product designers or practitioners to computationally design objects with special tactile properties requiring finite element methods (FEMs) or other specialized knowledge about physics.

Prior efforts on haptics partially solved these issues, most of which focused on large objects. For example, Hudin et al. and Wockel et al. superimposed multiple frequencies from multiple vibrators to render richer tactile sensations [4, 7]. Also, Dhiab et al. achieved a local vibration at the selected region of the device by suppressing the vibration near the vibrators with heavy components and structures [1]. These methods need large spaces to place many vibrators or heavy and bulky structures, and thus applying these methods to many everyday devices or objects is difficult. DFM Touch enables the presentation of local vibration on several everyday devices, such as a game controller by automatically designing a structure connecting local vibration regions and other regions using the technology of compliant mechanisms [5]. However, designing these structures is still difficult for product designers or practitioners because the direction of the connections between each structure significantly influences how vibrations occur within the structure, and they need to understand the vibration model to design this appropriately.

In this paper, we introduce TactPrint, a method that involves embedding a lattice structure around the area where we want to induce local vibrations and then optimizing this 3D structure to create a tactile display with the desired local vibration characteristics. With the TactPrint system, we can obtain structures with desired local vibration characteristics simply by inputting a 3D model, which can be fabricated with a 3D printer and combined with an actuator to create a tactile display easily. Our system was implemented using the commercial 3D CAD software system, Rhino and Grasshopper. First, the user gives a 3D mesh as input and interactively specifies the region for the local tactile sensation. Then, the system automatically generates a 3D model incorporating a lattice structure as shown in Figure 1. We can use the appropriate resonant frequency of the specified region by tuning the lattice structure. In this paper, we optimized the shape of the lattice structure and suitable material parameters from the input model, specified region, and the selected frequency using gradient descent. Also, to reduce the computational cost of the FEM, we introduced the homogenization method that approximates the physical properties of a microstructure. In this paper, we explain the design pipeline of our method and demonstrate the applications with multiple everyday devices and objects.

2 METHODS

2.1 Overview

In our method, as shown Figure 1, users input a 3D model they wish to configure as a haptic display, the local regions within its structure where they want to present haptic feedback, their surrounding area where they want to change structural shape to generate the local vibration, and the resonant frequency of that haptic presentation into

the system. Based on these design variables, our method replaces the surrounding area with a lattice structure to generate local vibration. In addition to this, our method computes shape parameters that should be satisfied by the structure embedded around the area where local haptic presentation is desired. This is achieved through gradient-based optimization, using the resonant frequencies and mode shapes obtained from finite element method-based modal analysis. Then, the 3D model is obtained.

2.2 Structural conversion for local vibration

We discuss the theoretical details of our method. After the user inputs, firstly, our method replaces surrounding regions of local regions with a sinusoidal-shaped lattice structure to generate local vibration region. When a part of a region is assigned a material or structure that is softer than used in the whole structure, the deformation of the soft material or structure's part becomes larger than other parts. Using this property, the surrounding regions deform significantly and prevent propagating the vibration that is applied to the local vibrate region. As the limitation of fabrication materials in current digital fabrication technologies, including 3D printing, we adopt a soft structure to control vibration propagation.

This study adopts a sinusoidal-shaped lattice microstructure. The lattice structure is defined by two parameters: the ratio of sinusoidal amplitude and scale of microstructure A and the ratio of diameter and scale of microstructure d .

2.3 Optimization of local vibration characteristics

Secondly, our method automatically optimizes the structural shape parameters of the microstructure to control local vibration characteristics using FEMs. In our optimization, we have adopted the resonant frequency and the corresponding modal shape as vibration characteristics and our method minimizes the squared errors of the user-desired vibration frequencies and the resonant frequencies corresponding to the modal shapes representing the local vibration. While the structurally converted surrounding area suppresses the vibration outside of the local vibration region at a certain frequency, in the case of the user-desired frequency vibration, specific areas outside the local vibration region still have the potential to vibrate significantly. This is influenced by various structural factors, such as the input model, the position of the local vibration region, and the shape of the embedded microstructure. Generally, it is known that these vibration characteristics can be represented by summing the modal shapes of the resonant frequencies with high contribution rates [2] in various structures. The displacement x can be represented using resonant frequencies with a frequency of external vibration is $\frac{\omega'}{2\pi}$:

$$x = U\xi = \sum_k u_k \xi_k \quad (1)$$

$$\xi_i = \frac{1}{\omega_i^2} \frac{\sum_j u_{i,j} f_j}{1 - (\frac{\omega'}{\omega_i})^2} (\sin(\omega' t) - \sin(\omega_i t)) \quad (2)$$

It denotes that $U = u_k$ is the modal shape corresponding to the resonant frequency ω_k , and f_k is the amplitude of external force at applied from the finite element k . As shown in Equation 1 and 2,

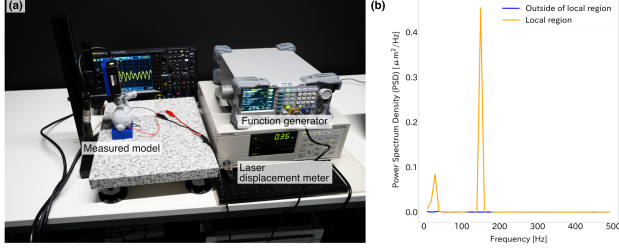


Figure 2: (a) Experimental setup of measuring vibration and (b) power spectrum density measured in the experiment. There are two measured points: the point where the user designed as a local region and the outside of the local region.

the structure vibrates like a modal shape when the applied vibration frequency is close to the corresponding resonant frequency. U includes the modal shape representing the local vibration of the user-desired region. Therefore, we have adopted the resonant frequency as a vibration characteristic.

The objective function follows as the function using the shape parameter of microstructure:

$$\underset{A,d}{\text{minimize}} L(A,d) = \sum_{i=1}^n \left(\frac{\omega_i(A,d) - \omega_{i,ref}}{\omega_{i,ref}} \right)^2 \quad (3)$$

It denotes that n is the number of local regions in the structure. These resonant frequencies and the corresponding modal shapes are obtained by modal analysis. Modal analysis solves the eigenvalue problem as follows, using the mass matrix $M(A,d)$ and stiffness matrix $K(A,d)$:

$$K(A,d)U = SM(A,d)U \quad (4)$$

Diagonal elements of S , $S_{i,i} = \omega_i^2$. The mass and stiffness matrices are represented as the function using the lattice shape parameters A and d . An increase of the parameter A , which represents the shape of beams constructing the microstructure, leads to a decrease in the stiffness of the microstructure and an increase in the mass, and vice versa. The stress experienced by winded beams under tension and compression is lower compared to straight beams. In addition, an increase of the parameter d , representing the thickness of the beams, results in an increase in both the stiffness and the mass.

Using the objective function, our method updates each lattice shape parameter as follows:

$$\begin{aligned} A^{(k+1)} &\leftarrow A^{(k)} - \eta \frac{L(A + \Delta A, d) - L(A, d)}{\Delta A} \\ d^{(k+1)} &\leftarrow d^{(k)} - \eta \frac{L(A, d + \Delta d) - L(A, d)}{\Delta d} \end{aligned}$$

According to the relationship between the shape of the microstructure and the physical properties, we adopt gradient descent optimization to minimize the objective function. Since it is difficult to solve the differential value of the discrete stiffness matrix analytically, we used a first-order approximation. The partial derivative of the eigenvalue problem using shape parameters can estimate the effect for the optimization. Using the sample shape parameters p , adaptable to d and A , the value obtained by differentiating the

j -th eigenvalue ω_j^2 is shown by the following equation:

$$\frac{\partial \omega_j^2}{\partial p} = \frac{u_j^T (-\lambda \frac{\partial M}{\partial p} + \frac{\partial K}{\partial p}) u_j}{u_j^T M u_j} \quad (5)$$

This equation implies that the resonant frequency increases when the stiffness matrix increases and the mass matrix decreases and becomes lower when the stiffness matrix decreases and the mass matrix increases.

In addition, we adopt the homogenization method to reduce calculation costs in calculating the mass and density matrices. The homogenization method computes the physical properties of repetition of microstructure as uniform material [3].

2.4 Implementation

We implemented this optimization using the 3D CAD software (Rhino 7 with Grasshopper component, Robert McNeel & Associates) and the CAE solver software (Ansys Mechanical, Ansys).

We fabricated the 3D model using a powder sintering 3D printer (Lisa Pro, Sinterit) and composed the haptic display using a linear resonant actuator (LD14-002, Nidec Copal).

3 EXPERIMENT AND RESULT

In this section, we conducted a study for this method. We verified the local vibration in the designed structure which optimized the resonant frequency of the local region as 150 Hz using Stanford Bunny model. The experimental setup is shown in Figure 2 (a). We placed this model on an anti-vibration table and applied a vibration of 150 Hz from the vibrator using a function generator (LG4222, IWATSU). Then, we measured the vibration at two points where the silver tape (No.AL-50BT, 3M) was affixed using a laser displacement meter (LC-2400, KEYENCE) with a laser head (LC-2440, KEYENCE).

The measurement result is shown in Figure 2 (b). The result of the measured vibration waveform and their power spectrum density (PSD) reveal that the vibration outside of the local region was suppressed compared with the local region. The PSD of the local region was $4.54 \times 10^{-1} \mu\text{m}^2/\text{Hz}$, and the outside point was $1.15 \times 10^{-4} \mu\text{m}^2/\text{Hz}$.



Figure 3: The smartphone case. (a) The overview. (b) A user holds it vertically. (c) When a user holds it horizontally, a user can use the additional vibrate regions to enhance the haptic experience.

4 APPLICATIONS

We introduce two application scenarios using haptic displays designed with our TactPrint system and show them in the demonstration. Figure 3 illustrates a smartphone case that enables selective haptic presentation to the fingers holding the smartphone horizontally by controlling the distribution of vibrations in the case.

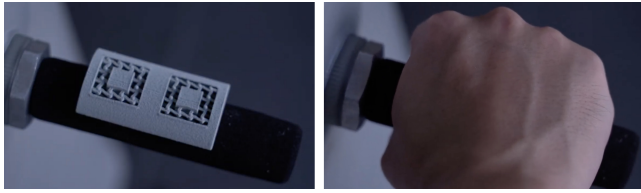


Figure 4: The navigate handle attachment of vehicles. (a) The overview. (b) A user grasp handle and drive a vehicle. The handle attachment allows navigation using a pseudo-motion sensation by controlling the vibration timing of two separated vibration areas.

The user can perceive vibrations in only the local region when the additional vibrator attached to the region is activated. In a user scenario utilizing this case, it can be used for feedback in multi-touch interactions, commonly occurring when holding a smartphone horizontally.

Figure 4 also shows a vehicle handlebar cover that provides a pseudo-motion sensation by controlling the vibration timing of two separated vibration areas within the cover, allowing for driving navigation using this feature. By vibrating the two separated vibration areas on the cover at appropriate timings, it is possible to induce a phantom movement of haptics [6], which can present a motion sensation to the user. Use scenarios could include providing notifications for right and left turning points or lane change points in conjunction with road navigation.

5 DISCUSSION

There are some limitations and lack of study about this method.

Frequency band control of local vibration. In the present study, local vibration is secured only at resonant frequencies, and the frequency band that realizes local vibration is not controlled. If the frequency band in which the local vibration occurs is controllable, the presentation position can be modified by the tactile stimulus (e.g., texture) being presented.

Resonant frequency matching of multiple mode shapes. Occasionally, vibrations at points other than the resonance point occur simultaneously. This phenomenon occurs when the resonance frequencies corresponding to each mode are close to each other when solving the eigenvalue problem, which is a simultaneous equation. If this phenomenon can be intentionally eliminated, it would be possible to reduce the number of design iterations the user requires.

Lack of parameter study of lattice structure. In this study, we only adopt the sinusoidal shape lattice structure. Adopting more lattice structures could make localized vibrations while maintaining a higher level of structural durability.

6 CONCLUSION

We have introduced TactPrint, a computer-based design method for 3D objects with local haptic presentation characteristics within their structure. We described the optimization methods of the system and demonstrated two application scenarios. In future work, we aim

to control the vibration frequency band by utilizing more complex microstructures and exploring application domains.

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